MKTG/STAT 4760/7760
Spring 2026

Homework 4

1. On Canvas you'll find a dataset reporting how many times each household (from a panel of 3035 participating members) purchased a particular brand of toothpaste out of their last five category purchases. Fit the beta-binomial model and its variants with (and without) spikes at p=0 (i.e., "hard-core never buyers") and/or p=1 ("hard-core always buyers").
   a. Which model would you choose as "best" and why?
   b. Using this model estimated on the given data, determine the implied penetration of the focal brand if the maximum number of purchases in the product category were actually 10?
   c. Derive and implement a "means and zeros" estimation scheme for the BB model (without any spikes) and apply it to this dataset.

2. Derive the posterior distribution for λ and the conditional expectation formula for an NBD model for a future period of length t^* as applied to a customer who made x purchases over a calibration period of length t.

3. Going back to the Billboard example from Session 2, consider the weekly exposure counts for two different customers. Using the same NBD parameters obtained in that session (r=.969, α=.218), calculate their posterior estimates for λ at the end of each week. Do the final estimates make sense? Briefly explain.

week	*1*	*2*	*3*
Johari	1	1	1
Fangyuan	3	0	0

4. Now consider a zero-inflated NBD model (i.e., an NBD with a "spike at zero") with parameters r, α, and π estimated on data from time period 1. Use Bayes Theorem to derive the probability that someone who made zero purchases is part of the "spike at zero" group. (Hint: you might want to look at the "spike at 1" paper on the Session 2A page.) Then, using this formula, derive the overall formula(s) for the expected number of purchases in time period 2 for a customer who made x purchases in time period 1.